

Average Waiting Time(AWT) = $(9+2+3+5+10) / 5 = 5.8 \text{ ms}$

(VII) Recourses Required

- Computer System with basic configuration. Linux/Ubuntu/ CentOS /any other open source Operating System.
- **Software** : Turbo C/ vi editor/ Any open source Python IDE such IDLE, Jupyter Notebook, Spyder, PyCharm etc.)

(VIII) Precautions to be followed

- (1) Handle computer system with care.
- (2) Strictly follow the instructions for writing, compiling, and executing the program.
- (3) Don't forget to save file before execution.
- (4) Follow safety Practices

(IX) Program Code

Implement the Round Robin (RR) CPU Scheduling Algorithm to calculate average waiting time and average turnaround time.

Note : Attach the code at the end.

(This practical can be performed in any of the compiler like Turbo C/ vi editor etc or Python Interpreter / IDE like any open source IDE such IDLE, Jupyter Notebook, Spyder, PyCharm etc.)

(X) Conclusion

Successfully implemented C/Python program to calculate average waiting time and Turnaround Time of n processes with Round Robin (RR) CPU scheduling algorithm.

(XI) Practical Related Questions

Note : Below given are few sample questions for reference. Teachers must design more such questions to ensure the achievement of identified CO.

- (1) What is time quantum?
- (2) State advantages and disadvantages of round robin scheduling.
- (3) How context switching occurs in Round Robin scheduling and its impact on system performance?
- (4) Consider the processes P1,P2 ,P3,P4, P5 given in the below table. Find average waiting time and average turnaround time by using Round Robin Scheduling algorithm. Time quantum=3ms

Process	Burst Time	Arrival Time
P1	8	0
P2	7	1
P3	2	5
P4	3	6
P5	5	8

[Space to Write for Answers]

1) Time quantum (or time slice) is the fixed amount of CPU time assigned to a process in Round Robin scheduling. If the process does not complete within this time, it is preempted and moved to the end of the ready queue, allowing the next process to execute.

2) Advantages :-

1. Fair CPU allocation to all processes.
2. Prevents starvation.
3. Provides good response time for interactive systems.
4. Easy to implement.

Disadvantages :-

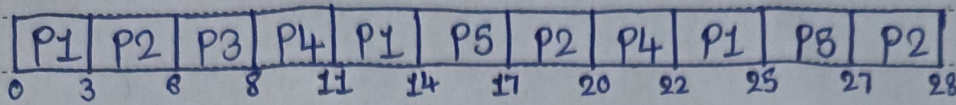
1. High context-switching overhead.
2. Performance depends on the choice of time quantum.
3. Can result in higher waiting and turnaround times.
4. Less efficient for long CPU-bound processes.

- 3) 1. A process is assigned the CPU for one time quantum.
2. If the process finishes within that time, the CPU is allocated to the next process.
3. If the process does not finish, the operating system :-
 - Saves the current process's state (registers, program counter, etc.)
 - Moves the process to the end of the ready queue.
 - Loads the state of the next process.
4. The CPU then starts executing the next process.

4) Given :- Time Quantum (TQ) = 3ms

Process	Arrival Time (AT)	Burst Time (BT)	CT	TAT	WT
P1	0	8	25	25	17
P2	1	7	28	27	20
P3	2	2	8	6	4
P4	3	5	22	19	14
P5	5	6	27	22	16

Gantt Chart :-



$$TAT = CT - AT$$

$$WT = TAT - BT$$

$$P1 = 25 - 0 = 25$$

$$P1 = 25 - 8 = 17$$

$$P2 = 28 - 1 = 27$$

$$P2 = 27 - 7 = 20$$

$$P3 = 8 - 2 = 6$$

$$P3 = 8 - 2 = 4$$

$$P4 = 22 - 3 = 19$$

$$P4 = 19 - 5 = 14$$

$$P5 = 27 - 6 = 22$$

$$P5 = 22 - 6 = 16$$

$$\text{Average Waiting Time: } \frac{17+20+4+14+16}{5} = \frac{71}{5} = 14.2 \text{ ms}$$

$$\text{Average Turnaround Time: } \frac{25+27+6+19+22}{5} = \frac{99}{5} = 19.8 \text{ ms}$$

Results:-

$$\text{Average Waiting Time} = 14.2 \text{ ms}$$

$$\text{Average Turnaround Time} = 19.8 \text{ ms}$$